

REIPA, Norway

WMO: 011140

Lat: 66.9033N

Lon: 13.6458E

Elev: 26

StdP: 14.68

Time Zone: 1.00 (EUC)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.1	11.8	0.3	5.6	12.5	3.2	6.5	14.4	40.6	41.8	33.7	41.7	4.8	60	0.699

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.3	73.1	60.1	69.2	58.5	66.1	57.1	62.5	71.1	59.9	67.0	58.1	64.2	5.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.4	73.0	66.1	56.6	68.3	63.1	54.9	64.1	61.1	28.1	71.3	26.4	67.2	25.2	64.3	70.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.3	27.3	23.0	5.0	78.2	2.9	4.0	2.8	81.1	1.1	83.4	-0.5	85.7	-2.6	88.6		
(5)			4.3	65.3	3.7	2.8	1.6	67.4	-0.5	69.0	-2.6	70.6	-5.3	72.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	40.9	28.2	28.9	31.4	37.3	45.2	50.2	56.2	55.2	50.0	41.4	35.1	30.8	
	DBStd	11.86	8.25	8.15	7.15	5.48	6.19	5.42	5.35	4.50	4.95	6.32	7.51	8.03	
	HDD50	3850	675	590	575	384	178	62	2	7	59	275	447	596	
	HDD65	8810	1140	1010	1040	832	615	446	282	305	449	733	897	1061	
	CDD50	527	0	0	0	2	28	67	194	169	60	7	0	0	
	CDD65	12	0	0	0	0	0	1	9	2	0	0	0	0	
	CDH74	103	0	0	0	0	1	9	83	10	0	0	0	0	
	CDH80	13	0	0	0	0	0	0	13	0	0	0	0	0	
Wind	WSAvg	8.2	8.0	8.7	9.9	7.9	8.1	8.2	7.2	6.8	7.3	8.5	8.5	9.5	
Precipitation	PrecAvg	66.6	5.9	5.2	5.0	4.1	4.0	4.0	4.3	5.2	7.6	8.4	6.9	7.0	
	PrecMax	81.4	14.3	9.4	8.5	7.3	5.9	8.1	9.0	13.9	14.3	15.6	13.4	12.9	
	PrecMin	53.5	1.0	2.8	1.7	1.0	1.5	0.9	1.1	1.4	1.9	2.4	0.9	0.9	
	PrecStd	7.5	3.4	1.9	2.1	1.6	1.5	1.9	2.1	2.6	3.1	3.5	3.3	3.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.3	45.6	48.8	62.6	72.2	74.4	81.6	75.1	67.3	61.5	51.9	48.1	
		MCWB	43.2	41.5	40.0	48.2	55.4	57.6	65.2	62.3	55.0	52.1	47.2	42.5	
	2%	DB	43.6	43.9	45.0	53.5	66.3	67.6	75.7	71.0	64.1	55.9	49.1	46.0	
		MCWB	40.2	40.4	40.2	44.2	53.3	56.0	63.0	60.2	53.9	48.6	44.5	42.8	
	5%	DB	41.6	42.4	43.4	48.1	60.6	63.9	70.1	68.1	61.0	52.9	47.4	44.1	
		MCWB	38.5	39.2	39.4	41.0	51.1	54.3	59.8	58.8	52.8	47.7	43.6	40.7	
	10%	DB	39.5	41.0	41.8	45.6	55.8	60.1	66.0	64.8	58.2	50.3	45.5	42.0	
		MCWB	36.4	37.8	38.0	40.0	48.0	53.0	58.8	57.2	52.4	46.2	41.9	38.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.1	43.2	43.1	49.3	56.3	59.8	67.3	64.2	57.7	52.9	48.1	45.4
			MCDB	45.7	44.5	44.8	60.9	68.6	69.6	77.9	71.4	63.6	60.9	49.7	46.6
2%		WB	41.0	41.4	41.5	45.0	54.1	57.6	63.8	61.5	55.6	50.2	45.9	43.6	
		MCDB	42.8	43.2	44.0	51.0	64.6	66.3	73.5	69.5	60.4	54.4	48.4	45.4	
5%		WB	38.8	39.9	39.9	42.6	51.4	55.3	61.3	59.4	54.3	48.4	44.2	40.7	
		MCDB	41.3	42.3	43.2	46.1	59.2	62.3	68.7	66.3	58.9	51.9	47.0	43.9	
10%		WB	36.8	37.7	38.1	41.1	48.7	53.4	58.9	57.8	53.0	46.7	42.1	38.5	
		MCDB	39.5	40.5	41.5	44.8	55.1	58.9	64.7	63.7	57.3	50.0	45.0	41.6	
Mean Daily Temperature Range		5% DB	MDBR	10.8	11.3	12.9	13.5	13.2	11.0	12.3	13.7	13.1	10.6	9.7	10.2
			MCDBR	10.0	10.5	11.1	17.8	22.4	19.5	22.0	20.3	17.1	14.2	10.7	10.3
	MCWBR		8.3	9.0	8.5	11.7	12.0	10.5	10.9	11.2	10.0	9.4	8.0	8.9	
	5% WB	MCDBR	9.6	9.0	9.6	14.3	20.8	16.3	19.1	18.6	14.6	11.2	9.6	9.7	
		MCWBR	8.8	8.6	8.2	10.4	11.8	9.2	10.1	11.0	9.4	8.2	8.4	8.7	
Clear-Sky Solar Irradiance	taub	0.368	0.290	0.321	0.343	0.347	0.344	0.352	0.340	0.327	0.298	0.350	N/A		
	taud	2.590	2.166	2.258	2.343	2.418	2.479	2.491	2.515	2.522	2.306	2.541	N/A		
	Ebn at Noon	57	169	230	258	270	274	268	259	234	168	51	0		
	Edh at Noon	5	16	25	30	30	30	28	25	20	14	5	0		
All-Sky Solar Radiation	RadAvg	15	121	482	1013	1460	1632	1464	1094	587	210	34	0		
	RadStd	2	21	66	98	100	209	152	108	66	41	4	0		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (22 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page